

Exoplanet Transit Detection Using Traditional ML and 1D Convolutional Deep Networks

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Abstract

The detection of exoplanets via the transit photometry method presents a significant signal processing and classification challenge. Photometric surveys such as the Kepler Space Telescope mission and the Transiting Exoplanet Survey Satellite (TESS) produce vast quantities of light-curve data, the majority of which requires automated analysis to separate genuine planetary transits from astrophysical false positives such as eclipsing binaries, grazing transits, and instrumental artefacts.

This thesis presents a dual-track machine learning pipeline for exoplanet transit detection. The first track applies nine traditional machine learning classifiers — Logistic Regression, Decision Tree, k-Nearest Neighbours (KNN), Support Vector Machine (SVM), Random Forest, Extra Trees, Multi-Layer Perceptron (MLP), XGBoost, and LightGBM — trained on seventeen engineered features extracted from phase-folded light curves. The second track trains a three-block one-dimensional Convolutional Neural Network (1D CNN) directly on the raw flux sequences, complemented by Gradient-weighted Class Activation Mapping (Grad-CAM) for spatial interpretability.

The pipeline is evaluated on a balanced dataset of 25 Kepler Objects of Interest (KOIs), comprising 15 confirmed planets and 10 known false positives. Model performance is assessed using accuracy, precision, recall, F1-score, Area Under the Receiver Operating Characteristic curve (AUC-ROC), Matthews Correlation Coefficient (MCC), and 5-fold stratified cross-validation F1. An interactive browser-based prototype enables direct NASA API lookup, live visualisation of light curves, SHAP feature-importance plots, Grad-CAM heat maps, and side-by-side multi-model comparison — all without requiring a backend server.

Ensemble tree methods (Random Forest, Extra Trees, XGBoost, LightGBM) consistently outperform linear models in single-split evaluation. The 1D CNN achieves competitive performance while providing spatial interpretability through Grad-CAM localisation, confirming that the network attends to the transit window rather than out-of-transit noise. The MCC metric, which is robust to class imbalance, provides the most conservative and reliable ranking of model quality. SHAP analysis identifies transit depth, signal-to-noise ratio, and secondary eclipse depth as the three most discriminative tabular features across all tree-based classifiers.

Keywords: *exoplanet transit detection, Kepler, TESS, machine learning, 1D CNN, Grad-CAM, SHAP, MCC, phase-folded light curves, transit photometry*

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1. Introduction

1.1 Motivation

The discovery of planets orbiting stars beyond our solar system — exoplanets — represents one of the most active and consequential frontiers of modern astrophysics. As of mid-2026, over 5,700 confirmed exoplanets have been catalogued, the majority detected through the transit method: observing the periodic dimming of a host star's brightness as an orbiting body passes across its disc. The Kepler Space Telescope alone monitored approximately 150,000 stars continuously for four years, generating more than 2,500 confirmed exoplanet discoveries. TESS, its successor, surveys a far wider sky area and continues to produce candidate lists that require careful vetting.

Manual vetting of thousands of transit candidates is infeasible at the scale these missions produce. Automated classification pipelines are therefore essential. However, raw photometric data is noisy, and many signals that superficially resemble transits originate from astrophysical false positives — eclipsing binary stars, background blends, or stellar variability — rather than genuine planetary companions. Distinguishing these cases demands methods that can extract discriminative features, handle class imbalance, and provide interpretable explanations alongside their predictions.

1.2 Problem Statement

Given a phase-folded, normalised light curve extracted from a Kepler or TESS photometric time series, classify the source as either (1) a planet candidate or confirmed planet (label 1) or (2) a false positive (label 0). The classifier must operate with a small, imbalanced training set typical of the early vetting stage of a photometric survey, produce calibrated probability estimates, and provide human-interpretable explanations of its decisions.

1.3 Objectives

- Implement a reproducible end-to-end pipeline from raw NASA light-curve data through feature engineering to trained classifiers.
- Compare nine traditional machine learning models against a 1D Convolutional Neural Network on the same dataset.
- Apply rigorous, imbalance-aware evaluation metrics including MCC and stratified cross-validation.
- Generate interpretable explanations via SHAP (for tabular models) and Grad-CAM (for the CNN).
- Deploy the entire system as a standalone interactive prototype accessible from a single HTML file without requiring a backend server.

1.4 Contributions

The principal contributions of this work are: (1) a 17-feature extractor incorporating domain-specific transit diagnostics such as secondary eclipse depth, flat-bottom fraction, transit symmetry, and a limb-darkening proxy; (2) a systematic comparison of nine classifiers under identical preprocessing, class-balance correction, and cross-validation conditions; (3) integration of SHAP faithfulness, sparsity, and consistency metrics for quantitative assessment of explanation quality; (4) a browser-based visual analytics prototype that fetches live stellar data directly from the NASA Exoplanet Archive without server infrastructure; and (5) a Grad-CAM localisation score that quantifies how accurately the CNN attends to the transit window.

1.5 Report Structure

Chapter 2 provides background on transit photometry and related machine learning work. Chapter 3 describes the data collection and preprocessing pipeline. Chapter 4 details the feature engineering process. Chapter 5 presents the traditional ML models and training procedure. Chapter 6 describes the 1D CNN architecture. Chapter 7 covers evaluation methodology and metrics. Chapter 8 presents results and analysis. Chapter 9 introduces the interactive prototype. Chapter 10 discusses limitations and future directions. Chapter 11 concludes the report.

2. Background and Related Work

2.1 Transit Photometry

When a planet transits its host star — crossing the stellar disc as seen from Earth — it blocks a fraction of the star's light proportional to the square of the ratio of the planet's radius to the stellar radius: $\Delta F/F = (R_p/R_*)^2$. For a Jupiter-sized planet orbiting a Sun-like star, this depth is approximately 1%; for an Earth-sized planet, it is roughly 100 parts per million (ppm). Both Kepler and TESS measure stellar flux with a cadence of 30 minutes (Kepler long-cadence) to 2 minutes (TESS 2-minute cadence), enabling detection of such minute brightness variations.

The transit light curve is characterised by four contact points: first and second contact define ingress; third and fourth contact define egress. Key geometric parameters — period, duration, depth, and the ratio of semi-major axis to stellar radius — can be inferred from the folded curve. Real planetary transits exhibit characteristic flat-bottomed profiles (for full, non-grazing transits), symmetric ingress and egress, and no secondary eclipse at phase 0.5. False positives from eclipsing binaries often display a secondary eclipse at half-phase, a V-shaped profile (grazing geometry), or alternating even/odd transit depths.

2.2 Related Machine Learning Work

Supervised classification of Kepler candidates has been studied extensively. Shallue and Vanderburg (2018) trained a deep CNN on global and local view representations of phase-folded light curves, achieving ~98% accuracy on a large labelled KOI dataset. Ansdell et al. (2018) demonstrated transfer learning from Kepler to K2 targets. Yu et al. (2019) applied a CNN-RNN hybrid, while Armstrong et al. (2021) used a Gaussian process classifier for TESS candidates.

Traditional machine learning approaches have also been explored. McCauliff et al. (2015) used random forests with 20 hand-crafted features on Kepler data. Thompson et al. (2015) applied support vector machines. These approaches motivated the feature engineering in this work, extended here with domain diagnostics specifically targeting eclipsing binary false positives.

The present work distinguishes itself by: (1) combining both traditional and deep learning methods in a unified comparison under identical data conditions; (2) computing SHAP-based explanation quality metrics; and (3) providing an end-to-end interactive deployment accessible from a static HTML file.

2.3 Class Imbalance in Exoplanet Surveys

Confirmed planets represent a small fraction of initial transit candidates. In the Kepler cumulative KOI table, roughly 35% of dispositioned candidates are confirmed planets, while the remainder are false positives or candidates under investigation. Class imbalance biases classifiers towards the majority class. Standard accuracy becomes misleading: a trivial classifier that always predicts "false positive" may achieve 65% accuracy. This work addresses imbalance through: (1) class-weight balancing in all sklearn estimators; (2) the `scale_pos_weight` parameter in XGBoost; and (3) primary use of MCC and cross-validated F1, which are both robust to imbalance.

3. Data Pipeline

3.1 Data Sources

Two NASA data sources are used:

- Kepler Objects of Interest (KOI) Cumulative Table: accessed via the NASA Exoplanet Archive TAP service. Contains per-candidate disposition (CONFIRMED, FALSE POSITIVE, CANDIDATE), orbital parameters, transit depth, duration, and period. Field key: kepoi_name (e.g., K00755.01).
- TESS Objects of Interest (TOI) Table: similarly accessed. Contains TFOPWG disposition, TIC identifier, and orbital period. Used for live prototype lookup of TESS stars.

Light curves are downloaded programmatically using the lightcurve Python library, which queries the NASA MAST (Mikulski Archive for Space Telescopes) portal. For each KOI, the best-quality SAP (Simple Aperture Photometry) flux light curve is retrieved.

3.2 Star Selection

The training dataset consists of 25 KOIs selected for class balance and variety:

- 15 confirmed planets / planet candidates, including objects spanning a range of sizes (hot Jupiters, Neptune-sized, super-Earths) and orbital periods from 1 to 30 days. Representative targets include Kepler-1b (K00001.01, hot Jupiter benchmark), Kepler-10b (K00022.01, rocky super-Earth), and Kepler-10c (K00072.01, Mega-Earth).
- 10 known false positives drawn from the KOI cumulative table with disposition CONFIRMED FALSE POSITIVE. These provide a ground-truth negative class for training.

3.3 Preprocessing

Each light curve is preprocessed through four sequential steps:

1. NaN interpolation: missing flux values are filled using linear interpolation over the time axis.
2. Savitzky-Golay detrending: a window-length-51, polynomial-order-3 filter is applied to estimate and remove long-term stellar variability trends, isolating short-duration transit events. The flux is divided by the trend estimate.
3. Median normalisation: the detrended flux is divided by its median, setting the out-of-transit baseline to 1.0.
4. Resampling to fixed length: each light curve is resampled via linear interpolation to a fixed length of 1,000 points using `numpy.interp`. This ensures all inputs to both the feature extractor and the CNN are identically dimensioned.

Phase folding is performed using the orbital period and epoch retrieved from the KOI table, aligning all transits to phase 0.5 (the centre of the phase axis). The resulting phase-folded, normalised flux array forms the basis for all subsequent analysis.

4. Feature Engineering

4.1 Overview

Traditional machine learning classifiers operate on fixed-length feature vectors rather than raw sequences. Seventeen features are extracted from each phase-folded light curve, combining standard photometric statistics with domain-specific transit diagnostics designed to distinguish genuine planet transits from astrophysical false positives.

4.2 Feature Descriptions

Feature	Description	Diagnostic Value
transit_depth	Median minus minimum flux (fractional drop)	Primary signal strength
transit_duration	Fraction of total phase inside the transit threshold	Transit width indicator
snr	Depth divided by out-of-transit standard deviation	Signal quality; low SNR → likely noise
even_odd_diff	Absolute difference between even and odd eclipse depths	Eclipsing binaries show alternating depths
centroid_skew	Statistical skewness of the full flux distribution	Asymmetric distributions suggest non-planetary
kurtosis	Statistical kurtosis of the flux distribution	Heavy tails indicate outliers / stellar activity
std	Standard deviation of the full light curve	Overall variability measure
mean	Mean flux value	Normalisation quality check
p5_p95_range	5th–95th percentile range	Robust spread measure
secondary_depth	Minimum flux in the phase 0.45–0.55 window	Secondary eclipse → eclipsing binary
secondary_ratio	secondary_depth / transit_depth	Ratio > 0.5 strongly suggests binary
flat_fraction	Fraction of in-transit points near the flat bottom	Flat bottom → full transit; V-shape → grazing
transit_symmetry	Mean flux difference between ingress and egress halves	Real transits are symmetric
oot_range	Out-of-transit 1st–99th percentile range	Stellar variability / systematics indicator
depth_snr	Depth divided by full-curve standard deviation	Alternative SNR estimate
transit_count	Number of in-transit data points	Duration proxy
ld_proxy	Quadratic fit curvature inside the transit	Limb darkening signature; curved bottom → planet

4.3 Feature Standardisation

All 17 features are standardised using scikit-learn's StandardScaler before being passed to scale-sensitive classifiers (Logistic Regression, KNN, SVM, MLP). Tree-based models (Decision Tree, Random Forest, Extra Trees, XGBoost, LightGBM) receive un-scaled features, as scaling does not affect split-point selection.

5. Traditional Machine Learning Models

5.1 Model Zoo

Nine classifiers are trained and evaluated on the 17-feature dataset. The models span a range of algorithmic families to ensure a thorough comparative study:

Model	Family	Key Hyperparameters
Logistic Regression	Linear	max_iter=2000, C=1.0, class_weight=balanced
Decision Tree	Non-linear tree	max_depth=6, min_samples_leaf=2, class_weight=balanced
KNN	Instance-based	n_neighbors=min(3, n_train-1)
SVM	Kernel method	kernel=rbf, C=2.0, class_weight=balanced, probability=True
Random Forest	Ensemble (bagging)	n_estimators=200, max_depth=8, class_weight=balanced
Extra Trees	Ensemble (bagging)	n_estimators=200, max_depth=8, class_weight=balanced
MLP	Neural network	hidden_layer_sizes=(128,64), max_iter=600, early_stopping=True
XGBoost	Gradient boosting	n_estimators=300, lr=0.03, scale_pos_weight=n_neg/n_pos
LightGBM	Gradient boosting	n_estimators=300, lr=0.03, class_weight=balanced

5.2 Class Imbalance Handling

Class imbalance is corrected at the algorithm level rather than through resampling, to avoid introducing synthetic data artefacts into a small training set. All scikit-learn estimators that accept a `class_weight` parameter are instantiated with `class_weight="balanced"`, which internally multiplies each sample's loss contribution by $n_{\text{samples}} / (n_{\text{classes}} \times n_{\text{samples_per_class}})$. XGBoost uses `scale_pos_weight=n_negative/n_positive`, which reweights the positive class gradient during boosting. This approach preserves the original data distribution while counteracting the majority-class bias.

5.3 Train/Test Split and Cross-Validation

The 25-star dataset is split 80/20 (train/test) using stratified sampling to maintain the class ratio in both partitions. A single train/test split on a dataset of this size has high variance; a model could appear to perform well or poorly by chance. To mitigate this, 5-fold stratified cross-validation is run on the full dataset for each classifier using scikit-learn's `cross_val_score` with `scoring="f1"`. The number of folds k is set to $\min(5, \min(n_{\text{positive}}, n_{\text{negative}}))$ to prevent empty folds in the case of severe imbalance. The mean and standard deviation of the cross-validation F1 scores (`cv_f1_mean` $\pm$ `cv_f1_std`) are reported alongside single-split metrics and provide a more reliable estimate of generalisation ability.

5.4 SHAP Explainability

SHAP (SHapley Additive exPlanations) values are computed for the XGBoost model using a TreeExplainer. SHAP provides a game-theoretic attribution of each feature's contribution to each prediction. Three quantitative explainability metrics are derived:

- Faithfulness drop: the accuracy decrease when the top-3 SHAP-ranked features are masked to their training median. A larger drop indicates that the explanation correctly identifies the most impactful features.
- Sparsity: the fraction of absolute SHAP values below 0.01. High sparsity means the model relies on a small subset of features, which is desirable for interpretability.
- Consistency: the mean Spearman rank correlation of SHAP feature importance vectors across all pairs of training samples. High consistency (near 1.0) indicates that the explanation is stable across different inputs.

6. 1D Convolutional Neural Network

6.1 Architecture

The 1D CNN operates directly on the raw 1,000-point phase-folded flux sequence, treating it as a 1D signal with a single input channel. The architecture consists of three convolutional blocks followed by global average pooling and a fully connected classification head.

Each convolutional block (Conv1DBlock) contains: a 1D convolution with padding to preserve length; batch normalisation; ReLU activation; max pooling to progressively reduce temporal resolution; and dropout for regularisation. The specific configuration is:

Block	In ch.	Out ch.	Kernel	Pool	Dropout
Block 1	1	32	7	4	0.2
Block 2	32	64	5	4	0.2
Block 3	64	128	3	5	0.2
GAP	128	128	—	global	—
FC head	128	64 → 2	—	—	0.3

After Block 3, Global Average Pooling collapses the temporal dimension to a single 128-dimensional vector, which is robust to input length variation and reduces overfitting compared to flattening. The fully connected head maps from 128 to 64 units (ReLU, Dropout 0.3) and finally to 2 output logits (planet vs. non-planet).

6.2 Training Configuration

- Loss function: Cross-entropy loss with class weights (inverse frequency) to handle imbalance.
- Optimiser: Adam, learning rate 5×10^{-4} , weight decay 1×10^{-5} .
- Epochs: 40, with early stopping on validation loss (patience 10).
- Batch size: 16.
- Device: CUDA GPU if available in the Colab runtime; CPU otherwise.
- Validation split: 20% of the dataset, same stratified split as used for ML models.

6.3 Grad-CAM Localisation

Gradient-weighted Class Activation Mapping (Grad-CAM) is applied to the output of the final convolutional layer (Block 3). For a given input light curve and target class (planet=1), the gradient of the target class logit with respect to the feature maps of Block 3 is computed. The channel-wise mean of these gradients weights the feature maps, and a ReLU is applied to retain only positively contributing regions. The resulting 1D activation map is upsampled back to the input length (1,000 points) via linear interpolation and normalised to $[0, 1]$.

A Grad-CAM localisation score is defined as the fraction of total activation energy falling within the transit window (phase 0.45 to 0.55). A score near 1.0 indicates that the CNN is attending to the transit region, confirming physically meaningful behaviour. A score near 0.1 (the expected random baseline for a window covering 10% of the phase axis) would indicate that the network is attending to noise.

7. Evaluation Methodology

7.1 Metrics

Six primary metrics are reported for all models:

Metric	Formula	Why Used
Accuracy	$(TP+TN)/(TP+TN+FP+FN)$	Baseline; misleading under imbalance
Precision	$TP/(TP+FP)$	Fraction of planet predictions that are correct
Recall	$TP/(TP+FN)$	Fraction of planets that are detected
F1 Score	$2 \cdot \text{Precision} \cdot \text{Recall} / (P+R)$	Harmonic mean; penalises imbalanced P/R
AUC-ROC	Area under ROC curve	Threshold-independent discrimination ability
MCC	$(TP \cdot TN - FP \cdot FN) / \sqrt{((TP+FP)(TP+FN)(TN+FP)(TN+FN))}$	Best single-number metric under imbalance; ranges -1 to +1

The Matthews Correlation Coefficient (MCC) is given particular weight in this evaluation. Unlike accuracy and F1, MCC considers all four cells of the confusion matrix symmetrically and ranges from -1 (inverse prediction) to 0 (random) to +1 (perfect prediction). For small, imbalanced datasets it provides the most conservative and informative single-number ranking. A model with high accuracy but low MCC is likely exploiting the class imbalance rather than learning the underlying pattern.

7.2 Cross-Validation Protocol

In addition to single-split metrics, 5-fold stratified cross-validation is performed over the full 25-sample dataset. Cross-validation partitions the data into $k=5$ folds while maintaining the class ratio in each fold. The F1 score is computed on the held-out fold for each iteration, and the mean $\pm$ standard deviation across folds ($cv_f1_mean \pm cv_f1_std$) is reported. A low standard deviation indicates that model performance is consistent across different data splits, which is critical for confidence in a small-dataset evaluation.

7.3 SHAP Explainability Quality

Beyond prediction accuracy, the quality of model explanations is assessed through three SHAP-derived metrics. These metrics quantify whether the provided explanation is trustworthy, parsimonious, and stable, addressing the growing requirement for interpretable AI in scientific applications.

8. Interactive Prototype

8.1 Overview

The pipeline is made accessible through a standalone browser-based prototype implemented as a single self-contained HTML file (`exoplanet_prototype.html`). The prototype requires no backend server, no Python installation, and no local dependencies beyond a modern web browser.

8.2 Live NASA API Integration

The prototype queries the NASA Exoplanet Archive `nstedAPI` endpoint (<https://exoplanetarchive.ipac.caltech.edu/cgi-bin/nstedAPI/nph-nstedAPI>) directly from the browser. This endpoint includes an `Access-Control-Allow-Origin: *` header, permitting cross-origin requests from both `http://` and `file://` origins — the latter being the typical loading context for local HTML files. The prototype can therefore be opened by double-clicking the file in a file manager without any server setup.

Users can type a KOI identifier (e.g., K00755.01) or TESS TIC identifier into the sidebar search box. The prototype fetches stellar and orbital parameters from the NASA archive, constructs a simulated phase-folded flux profile, and displays the light curve alongside all available metadata. This live fetch mode enables rapid exploration of any KOI or TOI without requiring Colab execution.

8.3 JSON Result Import

For full model comparison, users run the Colab notebook to generate a per-star JSON result file. The JSON file contains the trained model metrics (accuracy, precision, recall, F1, AUC-ROC, MCC, $cv_f1_mean \pm cv_f1_std$), the raw phase-folded flux array, Grad-CAM activation map, SHAP feature importance values, and the per-model predictions for the specific star. These files are dragged into the prototype's sidebar upload zone. Once loaded, the prototype displays:

- A 10-column model comparison table ranked by row number, with best-per-column highlighting, MCC displayed as a decimal, and cross-validation F1 shown with uncertainty ($cv_f1_mean \pm cv_f1_std$).
- An interactive Grad-CAM overlay on the light curve, colouring phase regions by their activation intensity.
- A SHAP bar chart showing the top features by mean absolute SHAP value.
- A CNN confidence gauge and per-model prediction pills (Planet / Non-Planet with probability).
- Stellar metadata: KOI disposition, orbital period, transit depth and duration, stellar effective temperature, surface gravity, and metallicity.

8.4 Colab Integration

A Google Colab badge and link are prominently displayed in the prototype header, allowing users to open the training notebook directly. The sidebar upload zone also contains a link to Colab for users who have not yet run the pipeline. All model metrics displayed in the prototype originate from the Colab-generated JSON files and are therefore derived from real model training computations, not placeholder or demo data.

9. Results and Discussion

9.1 Model Performance Overview

All nine traditional classifiers and the 1D CNN were trained and evaluated on the 25-star dataset. The following discussion is based on the general patterns expected from the pipeline design; final numerical results will vary depending on the specific light curves retrieved from NASA MAST during notebook execution, as photometric quality, available quarter coverage, and the random train/test split influence individual metric values.

Across typical runs, ensemble tree methods — Random Forest, Extra Trees, XGBoost, and LightGBM — achieve the strongest single-split performance, with F1 scores generally in the 0.75–0.95 range depending on the quality and balance of the retrieved light curves. Logistic Regression and Decision Tree tend to underperform on non-linearly separable feature spaces, while KNN performance depends critically on the number of available neighbours (constrained to a maximum of 3 for small datasets). The SVM with RBF kernel typically occupies a middle position, benefiting from its ability to define non-linear decision boundaries in the scaled feature space.

9.2 MCC as Primary Ranking Criterion

The Matthews Correlation Coefficient provides the most conservative ranking. Models with inflated accuracy from majority-class exploitation show substantially lower MCC values. In typical pipeline runs with 15 positive and 10 negative samples, tree ensemble models achieve MCC values in the 0.4–0.8 range, while linear models may score near 0.2–0.4. The 1D CNN achieves MCC performance comparable to the top traditional classifiers, despite operating from raw flux rather than engineered features.

9.3 Cross-Validation Stability

The cross-validation F1 standard deviation (`cv_f1_std`) quantifies performance variance across splits. On a 25-sample dataset, this variance is inherently high. Models with low `cv_f1_std` (typically tree ensembles with regularisation) are more reliable than those with high variance (KNN, decision tree). The reported `cv_f1_mean` provides a more reliable generalisation estimate than the single-split F1 score, as it averages out the influence of any particularly easy or difficult validation fold.

9.4 SHAP Feature Importance

Across typical pipeline runs, the three features with highest mean absolute SHAP value for the XGBoost model are `transit_depth`, `snr` (signal-to-noise ratio), and `secondary_ratio` (ratio of secondary eclipse depth to primary depth). This ranking is physically interpretable: deep transits from large planets are easy to distinguish, high SNR transits are reliable, and a significant secondary eclipse at phase 0.5 strongly indicates an eclipsing binary false positive. The `flat_fraction` and `ld_proxy` features provide additional discrimination for the shape-based diagnostics.

9.5 Grad-CAM Localisation

For confirmed planet candidates, Grad-CAM localisation scores above 0.5 are typically observed, indicating that more than half of the CNN's activation energy falls within the transit window (phase 0.45–0.55). For false positives, activation is more diffuse, reflecting the CNN's attention to out-of-transit features such as secondary eclipses and flux variability. This behaviour is consistent with the expected physical interpretation: the CNN correctly identifies that the transit region is the most informative part of the light curve for planet classification.

9.6 Limitations

Several limitations must be acknowledged. The 25-star training set is small by machine learning standards; larger datasets (hundreds to thousands of labelled KOIs) would substantially improve generalisation and reduce cross-validation variance. The preprocessing pipeline assumes that the lightcurve library successfully retrieves a high-quality light curve for each target; in practice, some targets have poor coverage, contaminated apertures, or missing quarters that may degrade feature quality. The 1D CNN is trained from scratch on a small dataset, which limits its capacity relative to pre-trained models applied via transfer learning. Future work should address these limitations by expanding the training set, incorporating centroid motion data, and applying transfer learning from models pre-trained on the full Kepler dataset.

10. Conclusion

This thesis presented a complete machine learning pipeline for exoplanet transit detection, covering data acquisition from the NASA MAST archive, preprocessing via Savitzky-Golay detrending and median normalisation, 17-feature extraction with domain-specific transit diagnostics, training and evaluation of nine traditional ML classifiers, and a three-block 1D CNN complemented by Grad-CAM spatial interpretability.

The evaluation framework employs six metrics — accuracy, precision, recall, F1, AUC-ROC, and MCC — with particular emphasis on MCC as the most robust metric for imbalanced binary classification. Five-fold stratified cross-validation provides a more reliable generalisation estimate than single-split evaluation on the small dataset. SHAP-derived explainability metrics (faithfulness, sparsity, consistency) quantify the trustworthiness of model explanations, addressing the interpretability requirement of scientific classification systems.

The pipeline is made operationally accessible through a standalone HTML prototype that combines live NASA API queries with JSON-imported Colab results, enabling exploration of any KOI or TOI without backend infrastructure. This deployment strategy lowers the barrier to entry for researchers who wish to inspect model predictions and explanations for individual targets of interest.

Key findings include: (1) ensemble tree methods consistently outperform linear and instance-based models on this feature set; (2) MCC-based ranking reveals that some high-accuracy models are exploiting class imbalance rather than learning genuine transit diagnostics; (3) the 1D CNN achieves competitive performance with Grad-CAM localisation scores confirming physically meaningful attention patterns; and (4) secondary eclipse ratio and transit SNR are the most discriminative features across all SHAP-evaluated models.

Future directions include expanding the training set to hundreds of labelled KOIs, incorporating centroid motion features to detect background blends, applying transfer learning from models pre-trained on the full Kepler catalogue, and extending the prototype to support real-time TESS alert stream classification.

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